**MSE 360: Kinetic Processes in Materials**
**Homework #4**

**Instructions**: Answer the following questions on a separate piece of paper. Be sure that your name and date are on the first page. Upload your work as a PDF to Moodle for credit. *All work must be done individually.*

1. An exact treatment of equilibrium reaction kinetics for reactions that do not go to completion was discussed in the dialog box in the text. Expressions 3.68 and 3.69 were provided as integrated rate laws for a simple equilibrium first-order reaction between A and B where the forward rate constant is given by $k_f$ and the backward rate constant is given by $k_r$. Prove that as $t \to \infty$, these expressions yield the equilibrium concentration of species A and B ($C_{Aeq}$ and $C_{Beq}$).